



Polyphasic Taxonomy of Species within the Dermatophyte Genus *Arthroderma* Including *A. fimbriatum* sp. nov and *A. cellare* sp. nov

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Abstract *Arthroderma* is a taxonomically diverse genus within the dermatophyte family (*Onygenales*, *Arthrodermataceae*) comprising primarily geophilic species typically found in animal burrows, where they feed on keratinous debris. In this study, species delineations within the genus were refined according to current taxonomic standards using the extensive array of *Arthroderma* strains of the BCCM/IHEM fungi collection. The methodology was based on a polyphasic approach including phenotypic analyses, multi-gene phylogenetic inference (comprising the internal transcribed spacer (ITS), β -tubulin (BT), and RNA polymerase II core subunit (RBP2) regions) and MALDI-TOF mass spectrometry. Special attention was given to the highly diverse *A. quadrifidum* clade which contains both geophilic and potentially

zoophilic species, assessing the phylogenetic position of recently described taxa within this group. In total, 77 strains from the BCCM/IHEM collection were (re-)examined. MALDI-TOF mass spectrometry was able to correctly identify 92% of the strains at species level. Additionally, certain isolates did not correspond to any previously described taxa, leading to the recognition, description and typification of two novel species.

Keywords *Arthroderma* · Dermatophytes · MALDI-TOF · Phylogenetics · *Arthroderma fimbriatum* · *Arthroderma cellare*

Introduction

The dermatophytes (*Onygenales*, *Arthrodermataceae*) are a group of fungi that are primarily characterized by their ability to metabolize and feed on keratinous tissues [1]. They can infect the nails, skin, and hair of animals and humans, causing diseases collectively known as dermatophytoses. Due to their pathogenicity, this group has attracted significant attention from both clinicians and taxonomists.

Throughout the twentieth century, numerous dermatophyte species were described based on phenotypic traits such as morphology, clinical symptoms, and physiology. This reliance on the phenotype led to over-classification of taxa. The advent of molecular techniques, along with the adoption of the

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one-fungus-one-name principle have significantly reduced the number of recognized species and enhanced our understanding of the classification within the *Arthrodermataceae* family [2–5]. A multi-locus phylogenetic analysis by de Hoog et al. reorganized the classification into seven genera: *Trichophyton*, *Epidermophyton*, *Nannizzia*, *Microsporum*, *Paraphyton*, *Lophophyton*, and *Arthroderma*, which have become widely accepted [6].

Despite significant progress, the taxonomy of the genus *Arthroderma* remains less resolved than that of other genera in the family, largely due to its predominantly geophilic nature. These taxa are often absent from clinical cases, limiting their representation in medical mycology studies and hindering comprehensive taxonomic evaluations. *Arthroderma* species are generally presumed to have their natural habitat in the burrows of animals or soil, where they feed on keratinous debris [7]. While they are occasionally found on the fur or skin of these animals, infections by species of *Arthroderma* are rarely observed. Notable exceptions include species within the *A. quadrifidum* clade; namely *A. redellii*, which causes infections on the skin of bats and the recently described species *A. mcgillisianum* and *A. lilyanum* which were originally isolated from domestic cats with dermatophytosis [8, 9]. It is currently unclear whether these species should be considered zoophilic or simply opportunistic geophiles [9], a distinction with important epidemiological and clinical implications. Classifying a species as zoophilic suggests a natural adaptation to animal hosts and a higher likelihood of causing infections, while opportunistic geophiles are typically environmental organisms that may only occasionally infect animals or humans under specific conditions [6].

Recent studies have provided insights into the phylogenetic relationships within the genus *Arthroderma*, based on analyses of the internal transcribed spacer (ITS), the β -tubulin (BT) and translation elongation factor 1-alpha (*tef1a*) gene regions [10, 11]. In addition, several new species have been described in recent years [9–11]. High intraspecies variability observed within species such as *A. quadrifidum* and *A. crocatum* indicates the possible existence of cryptic species [5]. Moreover, a recent survey of wild rodents resulted in the identification of four novel *Arthroderma* species, reinforcing the idea that this genus is likely more diverse than current estimates

suggest [11]. The BCCM/IHEM Fungi Collection holds a diverse selection of *Arthroderma* strains, representing a valuable resource for systematic mycological studies. Inclusion of these strains in a large-scale phylogenetic analysis can provide valuable insights into the composition and taxonomy of *Arthroderma*. Furthermore, accurate classification might influence diagnostic approach, epidemiological tracking and infection control strategy [6, 12, 13].

In this study, we employed a polyphasic approach, integrating multilocus phylogenetic analysis, MALDI-TOF mass spectrometry profiling, and morphological characterization, to revisit the taxonomy of the genus *Arthroderma*. Special emphasis is placed on cryptic lineages within the *A. quadrifidum* clade.

Materials and Methods

Isolation and Morphological Characterization

This study analyzed 77 strains from the BCCM/IHEM fungi collection, all classified under the genus *Arthroderma*. Detailed information regarding the origin and history of these strains is provided in Supplementary Table S1. The isolates were cultured on Sabouraud dextrose agar (S) and 1/10 diluted Sabouraud dextrose agar (S10) at both 25 °C and 37 °C for 7–21 days to assess their macroscopic features and growth rate. Medium S10 was selected to reduce nutrient availability, thereby slowing colony expansion and allowing more accurate measurement of growth rate. Since some isolates of the *A. quadrifidum* clade did not grow at 37 °C, this experiment was repeated at 12 °C for all isolates of this clade. Growth rate of isolates from the *A. quadrifidum* clade were determined under each of the aforementioned conditions by determining the area of growth at 7-day intervals.

Isolates of *A. fimbriatum* sp. nov. and *A. cellare* sp. nov. were additionally cultured on Potato dextrose agar (PDA) and Malt extract agar (MEA) and incubated for 7 and 14 days at 25 °C in the dark for macroscopic features. Colony colours were described using the colour chart by Kornerup & Wanscher (1978) [14]. Bromocresol Purple (BCP) agar and Christensen Urea liquid medium were incubated at 25 °C for 7 days to respectively test for alkalinity and urease action. Cultures on 1/10 diluted Sabouraud (S10)

were incubated for 12 days at 25 °C in the dark and were used to describe microscopic features. Slides were stained with Congo Red solution to increase visibility. Microscopic measurements were performed with a Nikon eclipse 600 microscope (Nikon Corp., Tokyo, Japan), using a 40× objective, and measured directly from the live image using NIS-Elements D software v. 5.30.01 (Nikon Corp., Tokyo, Japan). Microphotographs were obtained using a Leitz Diaplan microscope (Ernst Leitz GmbH, Wetzlar, Germany) operated in differential interference contrast (DIC) mode.

DNA Extraction, PCR, and Sequencing

Fungal strains were cultivated in Sabouraud dextrose broth for five days. For slow-growing isolates, incubation was extended until sufficient biomass was obtained. Genomic DNA was extracted using the Invisorb Spin Plant Mini Kit (Invitek, Berlin, Germany). Around 250 mg of fungal material was collected, frozen at −80 °C for 30 min and lyophilized overnight. This was followed by one minute of bead beating of the culture to enhance fungal cell wall disruption until a fine powder was reached. 400 µl lysis buffer and 20 µl proteinase K were used during lysis, and the lysis time was extended to 2–2.5 h. The mixture was vortexed and briefly centrifuged before being incubated for at least two hours at 65 °C. The mixture was then transferred onto the prefilter and centrifuged at 12,000 rpm, 200 µl of binding buffer was added and vortexed. The suspension was then transferred onto the spin filter and incubated for 1 min, then centrifuged at 12,000 rpm for 1 min and the filtrate was discarded. 550 µl of wash buffer was added and centrifuged at 12,000 rpm for 1 min, again the filtrate was discarded. This washing step was repeated 3 times, the final filtrate was discarded and the suspension was centrifuged for 2 min at 12,000 rpm. Finally 100 µl of prewarmed elution buffer was added and incubated for 3 min followed by 1 min of centrifugation at 10,000 rpm.

Two genomic regions were targeted for amplification and sequencing: (1) the partial β -tubulin (BT) gene using primers Bt2a and Bt2b as described by Glass and Donaldson [15], and (2) the internal transcribed spacer (ITS) regions using primers ITS5 and ITS4 [16]. The β -tubulin gene was selected as a secondary marker because of its high resolution in

delineating clades within dermatophyte taxa alongside ITS regions, the standard barcode for fungal identification [6, 17]. Although ITS and BT have proven informative for species-level identification, their phylogenetic resolution varies among taxa and the high variability of these markers may lead to unresolved intraspecific variation or alignment oversaturation at deeper nodes [10]. For more robust backbone inference, more conserved loci such as LSU, SSU, or *rpb2* are generally required. In this study, *rpb2* was additionally sequenced for the *A. quadrifidum* clade. For isolates in the *A. quadrifidum* clade the DNA-directed RNA polymerase II core subunit gene (*rpb2*) was sequenced as a tertiary marker using primers RPB2-7CF and RPB2-3053bR [18, 19]. All primers used can be found in the supplementary material.

PCR products were purified using the ExoSAP-IT PCR Product Cleanup kit (Affymetrix, Santa Clara, CA, USA) and Sanger sequenced using an ABI 3130xl Genetic Analyzer (Applied Biosystems, Waltham, MA, USA). Consensus sequences were assembled and edited using DNASTAR Lasergene 10 (DNASTAR, Madison, WI, USA). The resulting sequences were deposited in the European Nucleotide Archive (ENA) under the accession numbers available in the supplementary material (Table S1).

Phylogenetic analyses were conducted using BT and ITS sequences. Multiple sequence alignments were generated using MAFFT v7.505, employing the FFT-NS-i iterative refinement method and a 1PAM/ $\kappa=2$ scoring matrix. Alignments were manually inspected for inconsistencies. Using Partition-Finder v2.1.1, best fit model for partitioning were defined (see Supplementary Table S2) [20]. Maximum likelihood (ML) phylogenies were inferred with RaxML v8 [21] using 1000 bootstrap replicates to evaluate branch support. Bayesian inference was performed using MrBayes v3.2.7 [22]. For the *rpb2* dataset, sequences were aligned using MAFFT under the same settings as ITS and BT. Alignments were inspected manually. Intronic regions were retained and included in the alignment of the complete gene sequence.

MALDI-TOF Mass Spectrometry

For mass spectrometry analysis, the strains from the IHEM collection were cultured on Sabouraud Chloramphenicol (SC) agar plates at 25 °C for three days.

Incubation was extended to 7 days for slow-growing isolates, this included *A. uncinatum* strains IHEM 4793, IHEM 4796, IHEM 24388 and *A. ciferrii* IHEM 23564. For *A. redellii* IHEM 28938, growth time was extended to 14 days and incubation temperature was set at 12 °C. Strains selected for the reference database were cultured on five SC plates: four for main spectrum profile (MSP) generation and one for quality control (Table S3). For the identification panel, a single SC plate was used per isolate.

Protein extraction was performed following a modified version of the protocol by Normand et al. [23]. Briefly, the samples were suspended in 300 µL of deionized water, vortexed, and mixed with 900 µL of absolute ethanol. After pelleting and air drying, the samples were successively treated with 70% formic acid and acetonitrile (50 µL each), followed by vortexing and final centrifugation. Mass spectra were acquired using a Microflex LT MALDI-TOF mass spectrometer (Bruker Daltonics, Bremen, Germany) with default settings, and data were analyzed using MALDI Biotyper v2.1 software.

A reference MSP database was constructed from strains representing phylogenetically distinct species or clades, as determined by the phylogenetic analysis (see Supplementary Table). For each reference strain, four MSPs were generated, each by merging ten mass spectra from a single subculture. For the identification panel, four mass spectra replicates were obtained from the single culture of each isolate. Identification was considered valid when at least three out of the four replicates matched a species in the reference database with a log score ≥ 1.7 . When spectra of the examined strains were also present in the reference database these results were taken out of consideration. When only a single isolate per species was available, the MALDI-TOF method was thus impossible, preventing evaluation of *A. ciferrii*, *A. eboreum*, *A. gloriae*, *A. multifidum*, *A. silverae*, *A. thuringiense*, and *A. mcgillisianum*.

Results

Phylogenetic Analyses

In total, 35 species of *Arthroderma*, including the newly described *A. fimbriatum* and *A. cellare*, were included in the phylogenetic analyses. To our best

knowledge, this analysis contains all the currently known species included in the genus. The phylogenetic tree based on the ITS and BT multilocus alignment revealed five well-resolved clades supported both by high bootstrap scores and posterior probabilities in the maximum likelihood and Bayesian analyses respectively (Fig. 1). These clades were the *A. uncinatum*, *A. multifidum*, *A. silverae*, *A. terrestre*, and *A. quadrifidum* clades. Species represented by multiple isolates within these clades were all well supported and monophyletic, except for those within the *A. quadrifidum* clade. The species *A. amazonicum*, *A. chiloniense* and *A. crocatum* also formed a clade, however this clade received neither high bootstrap support nor high posterior probability (BS < 70 and PP < 0.90). Furthermore very long branch lengths indicate that these are distant lineages with unclear relationships to other species and clades within the genus. The exact phylogenetic placement of these species thus remains difficult to determine. The species *A. magnasporum* fell outside any of the described clades and can also be considered a distant lineage. The position of the clades and species was identical in both the maximum likelihood and Bayesian phylogenetic tree.

The *A. multifidum* clade harbored the highest diversity, comprising five well-supported species, namely *A. tuberculatum*, *A. phaseoliforme*, *A. cuniculi*, *A. multifidum*, and *A. eboreum*. Five additional taxa, *A. zoogenum*, *A. flavescens*, *A. vespertellii*, *A. melbournense* and strain IHEM 27270, each represented by a single strain, also formed separate lineages in the tree. IHEM 27270 is formally described here as *A. fimbriatum* sp. nov. Internal nodes also received high support in the Bayesian analysis, but in certain cases fell below 70 in the maximum likelihood analysis.

The *A. uncinatum* clade was a well-supported clade, with both internal nodes and species-defining nodes showing high support values in both analyses. All species represented by multiple isolates are monophyletic, while the species represented by a single isolate form distinct lineages. The following species are part of the *A. uncinatum*-clade: *A. uncinatum*, *A. lenticulare*, *A. gloriae*, *A. gertleri*, *A. insingulare*, and the recently described *A. rodenticum* and *A. simile*.

The *A. terrestre* clade received full support in both Bayesian and maximum likelihood analysis (BS = 100, PP = 1). The clade was also characterized

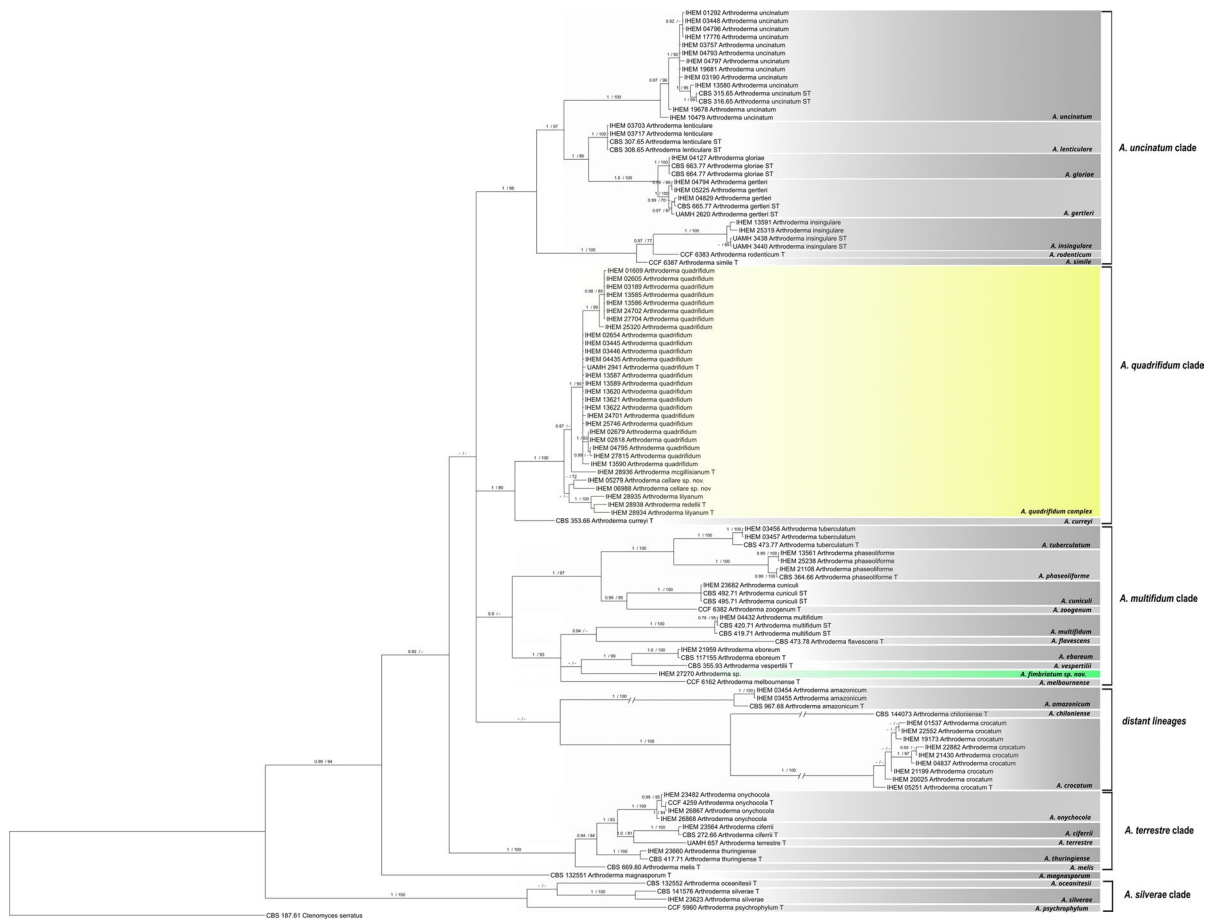


Fig. 1 Multilocus phylogenetic tree of the *Arthroderma* genus based on ITS and BT gene regions, with the type strain of *Ctenomyces serratus* (CBS 187.61) used as outgroup. Bayesian posterior probabilities and maximum likelihood bootstrap

values are shown at the nodes respectively. Bootstrap values lower than 75% and posterior probabilities lower than 0.9 are indicated with “–”

by high posterior probabilities and bootstrap values at internal nodes. Five species can be clearly distinguished, namely *A. onychocola*, *A. ciferrii*, *A. terrestre*, *A. thuringiense* and *A. melis*.

The most basally located clade in the phylogenetic tree was the *A. silverae* clade containing the species *A. oceanitesii*, *A. silverae* and *A. psychrophylum*.

The RPB2 gene was additionally sequenced and incorporated into the multilocus phylogenetic analysis of the *A. quadrifidum* clade, in order to enhance the resolution and support of the resulting phylogeny (Fig. 2). This marker was selected based on preliminary screening of commonly used fungal loci, which identified RPB2 as having high discriminatory power among species within the clade. The resulting

ITS+BT+RPB2 tree shows six distinct species, four of which (*A. quadrifidum*, *A. lilyanum*, *A. curreyi* and *A. cellare* sp. nov.) are well supported. The other two species, *A. mcgillisianum* and *A. redellii*, are each represented by a single strain, but appear as separate lineages in the tree.

MALDI-TOF Mass Spectrometry

Overall, 92% of the tested isolates were correctly identified at species level using MALDI-TOF mass spectrometry. In 7% of cases, no identification could be obtained. This was primarily due to either conflicting replicate results or low log scores (<1.7) among the top matches, which prevented reliable species

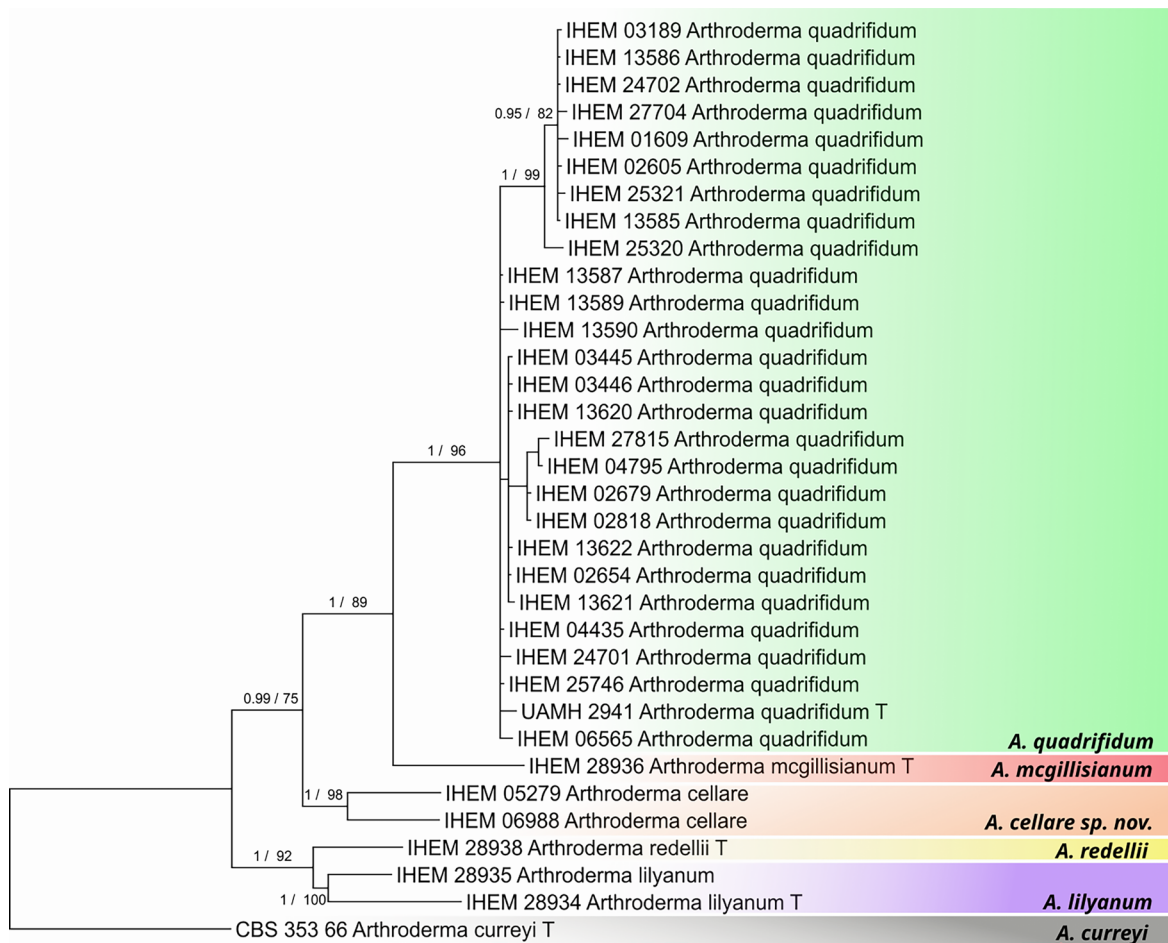


Fig. 2 Multilocus phylogenetic tree of the *A. quadrifidum* complex based on ITS, BT and RPB2 gene regions, with the type strain of *A. curreyi* (CBS 353.66) used as outgroup. Bayesian posterior probabilities and maximum likelihood boot-

strap values are shown at the nodes respectively. Bootstrap values lower than 75% and posterior probabilities lower than 0.9 are indicated with “—”

assignment. Specifically, no conclusive identification was possible for one isolate each of *Arthroderma onychocola*, *A. uncinatum*, *A. quadrifidum*, *A. cellare*, and *A. lilyanum*. In one instance, a misidentification occurred: the isolate IHEM 28395, representing *A. lilyanum*, was incorrectly assigned to the closely related species *A. quadrifidum*. Additionally, since *A. fimbriatum* sp. nov. is represented by a single isolate, its identification could not be evaluated, nonetheless it was not recognized as another species of the database. Out of the two *A. cellare* sp. nov. isolates, one was correctly identified, while the other yielded incongruent results across replicates and thus remained unclassified. While high identification accuracy was achieved for most clades, performance

notably declined when identifying members of the *A. quadrifidum* clade, particularly the rarer species (Table 1).

Growth Experiments

The growth rates of the species of the *Arthroderma quadrifidum* clade were explored under different temperature conditions and using different culture media (Table 2, Fig. 3). The Mann–Whitney U test was used to assess pairwise differences in fungal growth rates across different conditions. While it identified some significant differences between species, the results must be interpreted with caution due to small sample sizes in several groups. *Arthroderma*

Table 1 Species-level identification results of *Arthroderma* species using MALDI-TOF mass spectrometry. An identification was deemed valid if log scores were higher or equal to 1.7 and at least three out of the four replicates were identified as the same species. Strains for which these criteria were not met were counted as ‘no identification’. When identification did not match identification based on genetic markers it was deemed an ‘incorrect identification’

Clade/Species	Number of Samples			
	Total	Correct Identification	Incorrect Identification	No Identification
<i>Arthroderma amazonicum</i>	2	2 (100%)	0	0
<i>Arthroderma crocatum</i>	10	10 (100%)	0	0
<i>Arthroderma cuniculi</i>	2	2 (100%)	0	0
<i>Arthroderma gertleri</i>	3	3 (100%)	0	0
<i>Arthroderma insingulare</i>	2	2 (100%)	0	0
<i>Arthroderma lenticulare</i>	2	2 (100%)	0	0
<i>Arthroderma onychocola</i>	6	5 (83.3%)	0	1
<i>Arthroderma phaseoliforme</i>	3	3 (100%)	0	0
<i>Arthroderma tuberculatum</i>	2	2 (100%)	0	0
<i>Arthroderma uncinatum</i>	12	11 (91.6%)	0	1
<i>A. quadrifidum</i> clade				
<i>A. quadrifidum</i>	24	23 (95.8%)	0	1
<i>A. cellare</i>	2	1 (50%)	0	1
<i>A. lilyanum</i>	2	0 (0%)	1	1

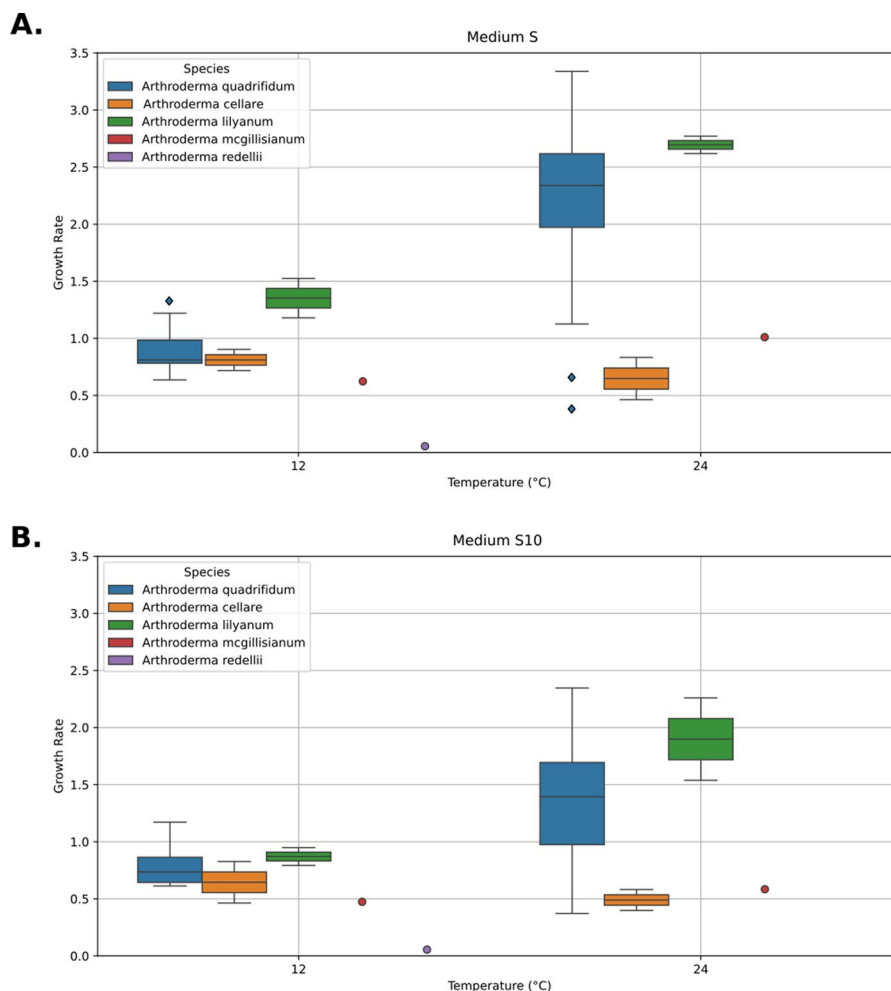
Table 2 Mean, median and variance of the growth rate in centimeter per day for species of the *A. quadrifidum* clade grown on the media S and S10 at temperatures 12 and 25 °C. No variance was calculated for *A. mcgillisianum* and *A. redellii* because only one strain was available. *A. redellii* showed no growth at 25 °C

Species	Medium	Temperature (°C)	Growth rate (cm/day)		
			mean	median	Var
<i>Arthroderma cellare</i> (n=2)	S	12	0,811	0,811	0,017
		25	0,648	0,648	0,068
	S10	12	0,645	0,645	0,066
		25	0,489	0,489	0,017
<i>Arthroderma lilyanum</i> (n=2)	S	12	1,352	1,352	0,059
		25	2,695	2,695	0,012
	S10	12	0,870	0,870	0,012
		25	1,899	1,899	0,261
<i>Arthroderma mcgillisianum</i> (n=1)	S	12	0,616	0,616	
		25	1,007	1,007	
	S10	12	0,466	0,466	
		25	0,578	0,578	
<i>Arthroderma quadrifidum</i> (n=22)	S	12	0,906	0,810	0,039
		25	2,153	2,339	0,514
	S10	12	0,784	0,736	0,028
		25	1,353	1,394	0,260
<i>Arthroderma redellii</i> (n=1)	S	12	0,035	0,035	
	S10	12	0,032	0,032	

quadrifidum exhibited the most pronounced temperature sensitivity, with significantly higher growth rates at 25 °C compared to 12 °C on both media S and S10 ($p < 0.01$ both) (Fig. 3). In contrast, *A. cellare* and *A. lilyanum* showed no significant differences in growth

rates between 12 °C and 25 °C, suggesting that these species are less affected by temperature variation within the tested range. Figure 3 illustrates the higher growth rate of *A. quadrifidum* at 25 °C compared to the newly described *A. cellare* as a potentially

Fig. 3 Boxplots comparing growth rate per day at 12 °C and 24 °C between species *A. quadrifidum*, *A. cellare* sp. nov., *A. lilyanum*, *A. mcgillisianum* and *Arthroderma redellii*; **A** Using Sabouraud medium (S) **B** Using 1/10 diluted Sabouraud medium (S10). Outliers, determined as datapoint further than 1.5 times the interquartile range, are indicated with an '◇'



useful distinguishing characteristic. Growth rates of the single strains available for *A. redellii* and *A. mcgillisianum* were also determined but since only one strain was available for each of these species, it is difficult to draw definitive conclusions. *A. redellii* did not grow at 25 °C, while also showing the slowest growth rate among all examined species at 12 °C using both media; 0.035 cm/day and 0.032 cm/day on S and S10 medium respectively. *A. mcgillisianum* had growth rates comparable to *A. cellare*, slightly lower at 12 °C and slightly higher at 25 °C (Fig. 4).

Taxonomy

Arthroderma fimbriatum F. Baert, E. D'hooge, P. Lefevere, sp. nov.

MYCOBANK MB859673. Figure 5a–f.

Typification: BELGIUM. WEST FLANDERS: Moorsele, from a horse hair sample, associated with hair loss, 16/03/2017, I. Meert (IHEM 27270). GenBank: ITS = OZ262013; BT = OZ262109.

Holotype: metabolically inactive (lyophilised) culture of strain IHEM 27270, deposited in the BCCM/IHEM culture collection.

Etymology: fimbriatum (Latin), referring to its fringed colony edge when grown in vitro.

Specimen examined: IHEM 27270 (ex-type culture); no other material available for examination.

Description: Colonies reaching 40 and 55 mm on potato dextrose agar (PDA) at 25 °C after respectively 7 and 14 days, downy to cottony. Colony color white to yellow-white (3A2). Reverse pale yellow (3A5) and olive brown (4E8) in the center. A dull yellow (3B4) pigment diffuses in the surrounding agar.

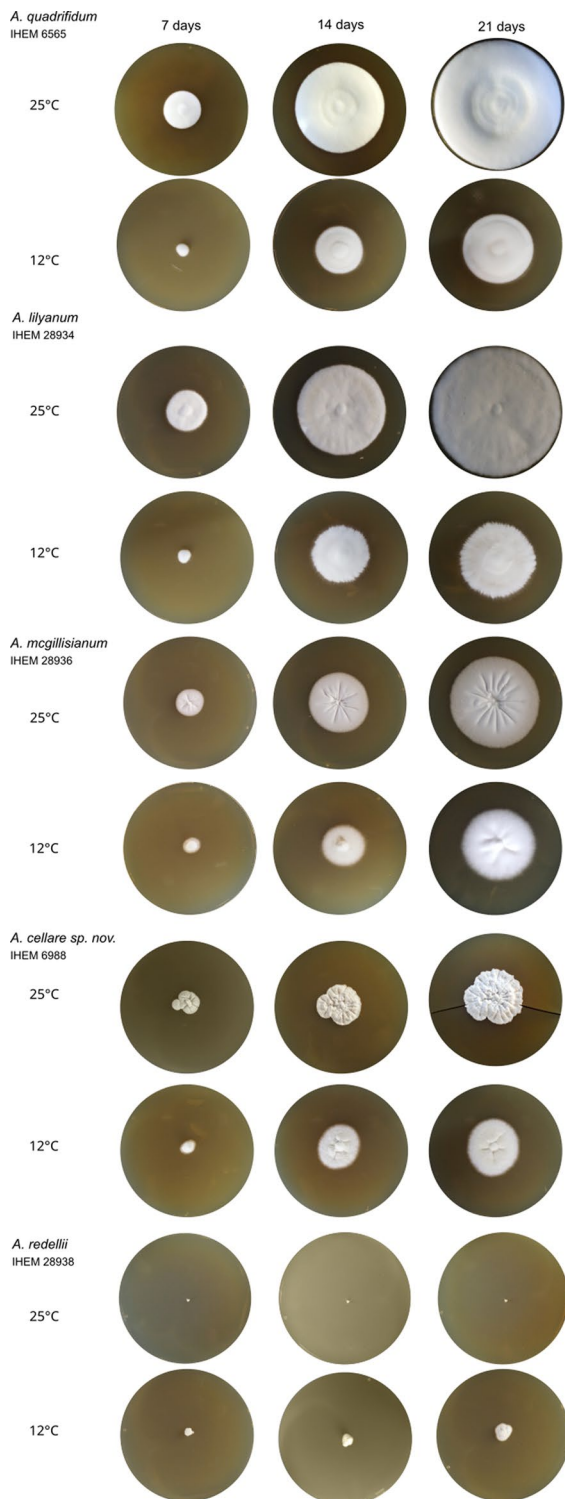


Fig. 4 Pictures illustrating the growth and micromorphology of species within the *Arthroderma quadrifidum* clade. Isolates shown here were grown on Sabouraud agar (S) for 21 days on both 25 °C and 12 °C

Colonies reaching 30 and 65 mm on malt extract agar (MEA) at 25 °C after respectively 7 and 14 days, loosely cottony and slightly granular, with a broad and submerged fringe. Aerial mycelium appears with age, in scattered, loose tufts. Colony color translucent to white, becoming greyish yellow (4B5) in the center with age. Reverse pale yellow (4A3) at the edge of the colony, changing to light yellow (4A5) and olive brown (4E8) in the center, becoming greyish yellow (4B6) overall and brown (5E8, 5F8) in the center with age. Colonies reaching 45 and 80 mm on Sabouraud dextrose agar (S) at 25 °C after respectively 7 and 14 days, cottony to downy with some granular spots. Colony edge fringed, colony surface flat to slightly sulcate. Clear exudate droplets are produced with age. Colony color white, becoming yellow-white (3A2) in the center with age. Reverse pale-yellow (2A4) at the edge of the colony, changing to yellow (2A7), bright yellow (3A8), and dark yellow (4A8) towards the center, becoming orange brown to brown (6C8, 5E8) in the center with age. A greyish yellow (4B6) pigment diffuses in the agar. Microconidia abundant, smooth-walled, formed directly on the hyphae or on short or longer stalks, or on side branches developing at acute angles. Microconidia mostly clavate, sometimes slender clavate or cylindrical, $(2.9) 3.2\text{--}4.5$ $(5.3) \times (1.7) 2.0\text{--}2.7$ $(3.3) \mu\text{m}$ (av. $\pm$ SD = $3.8 \pm 0.43 \times 2.3 \pm 0.25$, $n=50$), length to width ratio 1.3–2.1 (2.4) (av. $\pm$ SD = 1.7 ± 0.23). Hyphae can be constricted at the septa, giving them a knobby appearance. Nodular bodies and racquet hyphae occasionally present.

Alkaline reaction within 7 days on bromocresol purple lactose agar (BCP), 25 °C. Christensen liquid urea medium positive after 7 days, 25 °C. Growth on S + 3% NaCl, 25 °C. Good growth on S at 30 °C, no growth at 37 °C.

Notes/distinguishing characteristics: *Arthroderma fimbriatum* differs from closely related species by a combination of morphological and cultural characteristics, including abundant, smooth-walled and predominantly clavate microconidia, cottony to slightly granular colonies that are white to pale yellow with a fringed margin, absence of growth at 37 °C, and production of a yellowish pigment on most culture media. *Arthroderma vespertili* has larger microconidia ($5\text{--}8 \times 2\text{--}4 \mu\text{m}$) that become warty with age, the hyphae also become warty with age and produce sterile appendages (Vidal et al. 1996). *Arthroderma eboreum* produces cleistothecium-like structures

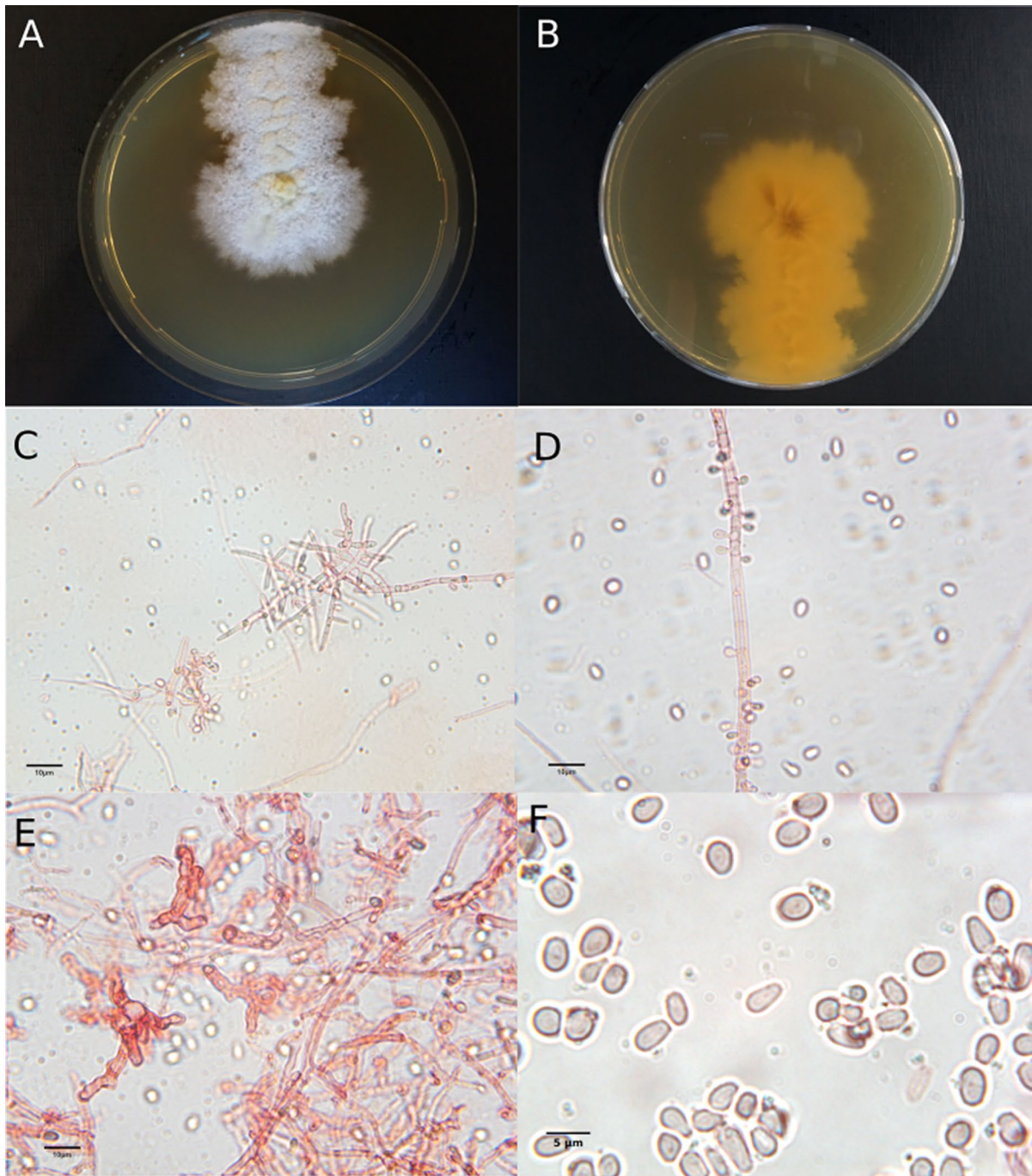


Fig. 5 *Arthroderma fimbriatum* IHEM 27270. **A–B** Front and reverse colony on S after 14 days at 25 °C; **C–D** Conidiophores 40×; **E** Nodular bodies 40×; **F** Microconidia 100×

and both microconidia ($4\text{--}5 \times 2.5\text{ }\mu\text{m}$) and macroconidia ($6\text{--}50 \times 4\text{ }\mu\text{m}$) (Brasch & Gräser 2005). *Arthroderma melbournense* produces coarsely granular

colonies and the microconidia have a length to width ratio that is slightly higher (1.3–3.4) than those of *A. fimbriatum* [10]. *Arthroderma flavescens* only

produces macroconidia and *Arthroderma multifidum* has larger, subglobose to obovate microconidia ($8.8\text{--}19.8 \times 6.6\text{--}13.2\text{ }\mu\text{m}$), occasionally with 1–2 septa [24, 25].

Arthroderma cellare F. Baert, E. D'hooge, P. Lefevere, sp. nov.

MYCOBANK MB859675. Figure 6a–f.

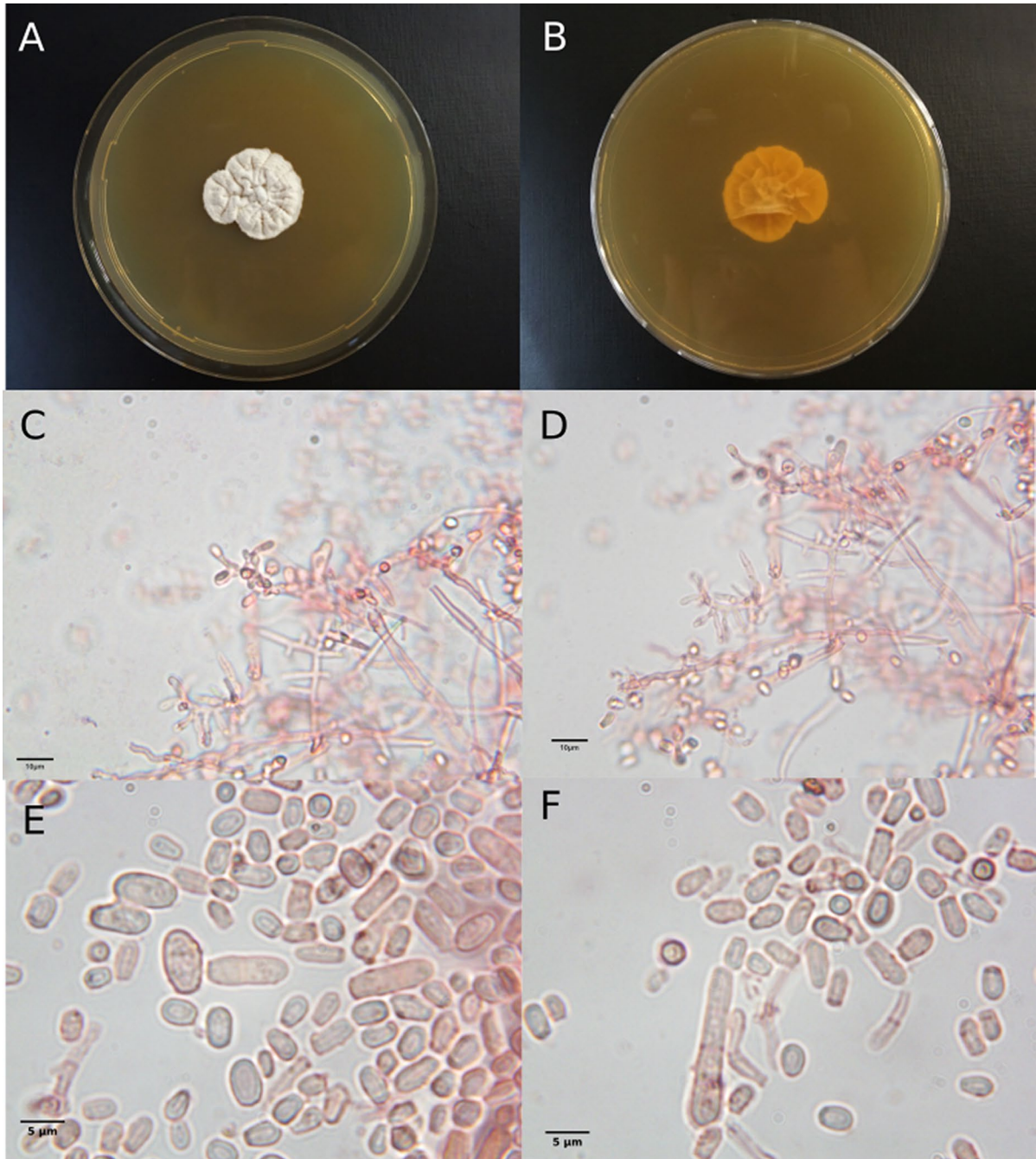


Fig. 6 *Arthroderma* IHEM 6988/IHEM 5279. **A–B** front and reverse colony on S after 14 days at 25 °C (IHEM 6988); **C–D** conidiophore 40x (IHEM 5279); **E–F** microconidia and intermediate conidia 100x (IHEM 5279)

Typification: IHEM 6988, isolated from air in a cellar in Ham-sur-Sambre, Namur, BELGIUM, isolated in 1992 by BCCM/IHEM, deposited in BCCM/IHEM. GenBank: ITS = OZ262047; BT = OZ262143; RBP2 = OZ262462.

Additional examined material: IHEM 5279, isolated from air in a cave in Folx-les-Caves, Walloon Brabant, BELGIUM, isolated in 1990 by BCCM/IHEM, deposited in BCCM/IHEM.

Holotype: metabolically inactive (lyophilised) culture of strain IHEM 6988, deposited in the BCCM/IHEM culture collection.

Etymology: cellare (Latin), referring to the habitat (cellar) where it was found.

Specimens examined: IHEM 6988 (ex-type culture) and IHEM 5279.

Description: Colonies reaching 16–18 mm and 31–35 mm on potato dextrose agar (PDA) at 25 °C after respectively 7 and 14 days, flat to downy with a granular texture where sporulating, can become slightly sulcate with age. Colony color white to yellow-white (3A2). Reverse yellow-white (4A2) at the edge of the colony, changing to grayish-orange (6B5) and brownish-orange (7C7–7C8) or reddish-brown (8D7) in the center. Colonies reaching 18–22 mm and 32–35 mm on malt extract agar (MEA) at 25 °C after respectively 7 and 14 days, flat with a granular texture where sporulating, tufts of downy aerial mycelium can appear with age. Colony color hyaline at the margin and white to yellow-white (3A2) where sporulating. Reverse hyaline to orange-white (5A2) at the edge of the colony, changing to light orange (5A4) and golden yellow (5B7–5B8) in the center. Colonies reaching 18–19 mm and 33–34 mm on sabouraud dextrose agar (S) at 25 °C after respectively 7 and 14 days, flat to downy with a granular texture where sporulating, can become slightly sulcate with age. Colony color white to yellow-white (3A2). Reverse yellowish-white (4A2) at the edge of the colony, changing to grayish-yellow (4A4–4B4) towards the center. Microconidia abundant, smooth-walled, formed directly on the hyphae or on short or longer stalks. Terminal clusters of conidia-bearing branches are common. These branches develop at acute angles. Microconidia quite variable in size, mostly clavate, sometimes slender clavate to cylindrical and sometimes inflated, $(2.9) 3.1\text{--}5.8 (6.2) \times (1.6) 1.8\text{--}2.7 (3.1) \mu\text{m}$ (av. $\pm$ SD = $3.9 \pm 0.69 \times 2.3 \pm 0.2$ 5, $n=50$), length to width ratio $(1.2) 1.4\text{--}2.8 (3.5)$

(av. $\pm$ SD = 1.7 ± 0.40 , $n=50$). In strain IHEM 5279, larger conidia are occasionally observed that exceed the size of typical microconidia and approach the lower size range of dermatophyte macroconidia. These conidia are cylindrical, ellipsoidal or club-shaped, up to 21 μm long, and may be one- or two-celled. Macroconidia are not observed. Racquet hyphae are occasionally present, hyphal spirals are abundant in older cultures.

Alkaline reaction within 7 days on bromocresol purple lactose agar (BCP), 25 °C, strong growth with a clearing underneath the colony. Christensen liquid urea medium positive after 7 days, 25 °C. Good growth on S + 3% NaCl, 25 °C. Good growth on S at 30 °C, no growth at 37 °C.

Notes/distinguishing characteristics: *Arthroderma cellare* can be recognized by a combination of morphological and cultural characteristics, including abundant clavate microconidia that are variable in size, the presence of spiral hyphae in older cultures, granular to downy colonies that are white to yellowish white, and absence of growth at 37 °C. *Arthroderma lilyanum* has cottony, white colonies and produces slightly smaller conidia ($2.3\text{--}4.9 \times 1.4\text{--}2.1 \mu\text{m}$) [9]. *Arthroderma mcgillisianum* has cottony and white colonies, produces larger conidia ($3.8\text{--}9.6 \times 3.1\text{--}3.4 \mu\text{m}$) and does not produce spiral hyphae [9]. The colonies of *A. redellii* are velutinous and do not grow at 25 °C [8]. This latter species produces microconidia, macroconidia and intermediate forms that range from 6 to 65 μm in length.

Discussion

Our phylogenetic analyses, based on concatenated sequences of the ITS and BT regions support the presence of 35 distinct *Arthroderma* species, distributed across five well-supported clades. The structure and composition of these clades is congruent with previously published phylogenies such as those of Hainsworth et al. and Moulikova et al. [10, 11]. Among these 35 species, two are newly described in this study, namely *A. fimbriatum* and *A. cellare*. *Arthroderma fimbriatum* is phylogenetically placed within the *A. multifidum* clade, alongside *A. eboreum*, *A. melbournense*, *A. vespertilii*, *A. flavescens* and *A. multifidum*. *Arthroderma cellare* is part of the *A. quadrifidum* clade which otherwise consists of four

closely related species, *A. quadrifidum*, *A. redellii*, *A. lilyanum* and *A. mcgillisianum*. Species delimitation within the *A. quadrifidum* clade proved challenging when relying solely on ITS and BT markers, as these failed to resolve all species boundaries with strong support. To address this, RPB2 was incorporated as a third genetic marker, significantly improving resolution. This resulted in a robust phylogeny with high bootstrap and posterior probability values for each of the five species in the clade. Although our phylogeny resolves *A. quadrifidum*, *A. lilyanum*, *A. mcgillisianum*, *A. redellii*, and *A. cellare* as well-supported lineages we recognize that current sampling remains uneven and geographically limited. We therefore regard these species boundaries as provisional pending analysis of larger, more geographically diverse sets of isolates. Previously defined species boundaries remained robust even with a relatively large number of isolates per species, for instance species pairs such as *A. gloriae* and *A. gertleri* or *A. tuberculatum* and *A. phaseoliforme* were clearly distinct and strongly-supported species in the phylogenetic tree [10, 11]. Interspecific variability consistently exceeded intraspecific variability across the markers used, enabling correct species identification and supporting further use of these markers in future taxonomic studies and clinical identifications when dealing with the genus *Arthroderma*.

The use of MALDI-TOF mass spectrometry demonstrated high accuracy in identifying most *Arthroderma* species, with 92% of isolates being correctly identified at species level. This is in line with the identification accuracy of previous studies focusing mostly on other, more clinically relevant genera within the *Arthrodermataceae* family [26–29]. Accuracy declined for species within the *A. quadrifidum* clade, likely due to limited availability of reference spectra and close phylogenetic relationship between them. In fact, since only one isolate of *A. mcgillisianum* and *A. redellii* was available, no identification was even possible for these species, as there were no reference spectra to compare to. The misidentification of one out of two *A. lilyanum* strains as *A. quadrifidum* underscores the need to expand reference databases to include newly described taxa. Future studies should prioritize the inclusion of additional reference isolates when available, to enhance MALDI-TOF MS reliability for rare and potential cryptic species. Having a wide diversity of spectra available in the

reference database is especially important when dealing with species complexes.

While *Arthroderma* species are predominantly geophilic, our findings highlight their potential for opportunistic infections, mainly within the *A. quadrifidum* clade. Species *A. mcgillisianum* and *A. lilyanum* are the causative agents of dermatophytosis in cats [9], furthermore *A. lilyanum* has been subsequently isolated from a lesion on the paw of a rabbit showing typical signs of dermatophytosis [30]. This raises questions about their zoonotic potential and that of other taxa within the genus. The same can be said for *A. redellii*, which causes dermatophytosis in hibernating bats [8]. Further studies are needed to clarify whether these species are truly zoophilic or merely opportunistic geophiles that occasionally infect mammals. Although they have not yet been identified in environmental studies, it is certainly possible that species like *A. redellii*, *A. mcgillisianum* and *A. lilyanum* are still primarily geophilic in nature as it has been reported for all the other species of *Arthroderma* known to date. Additionally, the discovery of *A. cellare* in subterranean environments suggests that caves and cellars may serve as underrecognized reservoirs for dermatophytes. A particularly intriguing observation is the apparent adaptation of *A. cellare* to lower temperatures. Unlike *A. quadrifidum*, *A. cellare* exhibited comparable growth at 12 °C and 25 °C, which suggests that *A. cellare* is better adapted to growing in colder subterranean environments such as caves and cellars. These observations for *A. cellare* are significant, but should still be interpreted with caution due to the low sample size, which may limit the generalizability of these findings. This apparent cold tolerance echoes the behavior of the closely related *A. redellii*, which can infect hibernating bats under low-temperature conditions. While no infections by *A. cellare* have yet been reported, its ecological niche and growth profile raise questions about its potential to colonize or infect cold-blooded or hibernating hosts. Interestingly, a study conducted in the Czech Republic revealed a large diversity of *Arthroderma* specimens in rodent hosts, including *A. quadrifidum*, *A. curreyi*, *A. rodenticum*, *A. simile*, *A. zoogenum*, *A. cuniculi* and *A. psychrophilum* [11]. Previous screening of bats however did not yield a high occurrence of *Arthroderma* species [31, 32]. On the American continent Vanderwolf et al.

(2015) identified *Arthroderma silverae* one single time on bat skin while Johnson et al. found no evidence of *Arthroderma* when screening 30 bats on five different locations [31, 32]. Screening of bat caves in Belgium yielded one single occurrence of *Arthroderma onychocola* [33]. These findings suggest that *Arthroderma* infection in these environments is either quite rare or missed by conventional sampling techniques. More targeted surveillance of dermatophytes on cave-dwelling mammals could help clarify whether *A. cellare* is an unrecognized agent of dermatophytosis or perhaps a non-pathogenic commensal in wildlife. Combining this surveillance with environmental air sampling of these habitats could provide valuable insights into the ecology of *Arthroderma* as well.

The discovery of *A. cellare* and *A. fimbriatum* further supports the notion that *Arthroderma* diversity remains underexplored, particularly in non-clinical settings. These novel taxa, originally identified from air samples in a cellar/cave and horse hair respectively, highlight the value of environmental sampling in broadening our understanding of fungal biodiversity, as well as the importance of re-evaluation of historically collected strains using modern biomolecular techniques.

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Data Availability Genetic sequences generated in this study were deposited in the ENA database under the accession numbers found in Supplementary Table S1.

Declarations

Conflicts of Interest The authors have no relevant financial or non-financial interests to disclose.

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